

The Fertility Summary Reports is a quick overview of both fertility-related genetics articles as well as some detoxification articles, which could tangentially link to fertility. Please see the linked articles for details and complete references.

🔗 Article: Infertility (Men)					
Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CFTR	rs113993960	--	D	0.003	DI – one copy of the Delta F508 cystic fibrosis mutation
PRM1	rs2301365	--	T	0.23	Increased risk of male infertility - DNA damage-response gene
LIG4	rs1805388	GG	A	0.15	Increased risk of male infertility - DNA damage-response gene
TGFB3	rs2284792	--	G	0.23	Increased risk of male infertility- chronic inflammation related
GSTM1	rs366631	--	A	0.78	AA: Increased risk of male infertility- chronic inflammation
HSPA1L	rs2227956	AA	G	0.15	Increased risk of male infertility; heat shock protein
NOS3	rs2070744	TT	C	0.34	CC: Increased risk of male infertility - nitric oxide synthase
MTHFR	rs1801133	GG	A	0.33	Increased risk of male infertility- folate related
MTRR	rs1801394	GA	G	0.5	Increased risk of male infertility; B12 related (B-complex)
FSHR	rs6165	--	T	0.52	Increased risk of male infertility; check hormone levels
FSHR	rs6166	TC	C	0.44	C/C Lower risk of male infertility; C/T: typical; TT: Slightly higher risk of male infertility; FSH hormone
HLA-DRA	rs3129878	CC	C	0.29	Increased risk of male infertility; Immune system
HFE	rs1800562	--	A	0.05	Increased risk of sperm damage; Iron overload related
HFE	rs1799945	CC	G	0.13	Increased risk of low sperm motility; Iron overload related
CFTR	i3000001	--	D	0.003	DI – one copy of the Delta F508 cystic fibrosis mutation
MTHFR	rs1801131	GT	G	0.3	Increased risk of male infertility- folate related

[Article: Infertility \(Women\)](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
MTHFR	rs1801133	GG	A	0.33	MTHFR C677T; reduced folate enzyme efficiency
MTHFR	rs1801131	GT	G	0.3	MTHFR A1298C; reduced folate enzyme efficiency
F5	rs6025	CC	T	0.02	Factor V Leiden; increased clot risk and miscarriage risk
F2	rs1799963	GG	A	0.01	Prothrombin variant; increased risk of clots and miscarriage
LHCGR	rs13405728	AG	G	0.08	Luteinizing hormone and hCG receptor; Increased risk of PCOS
LHCGR	rs2293275	TC	T	0.41	Luteinizing hormone and hCG receptor; Increased risk of PCOS
DENND1A	rs10818854	GG	A	0.04	Increased androgen synthesis; Increased risk of PCOS
FSHB	rs11031006	--	A	0.13	Increased lutenizing hormone to FSH ratio
FSHR	rs6166	TC	C	0.44	Folicle-stimulating hormone receptor; Increased PCOS risk
ADIPOQ	rs2241766	TT	G	0.1	lower risk of PCOS (good); T/T has higher risk of PCOS which can impact fertility
ADIPOQ	rs1501299	TG	T	0.25	TT: Decreased risk of PCOS (good)
MTNR1B	rs10830963	CC	G	0.27	Melatonin receptor in pancreas; increased risk of PCOS and higher insulin levels
F2	i3002432	--	A	0.001	Prothrombin variant; increased risk of clots and miscarriage

[Article: Inflammation in Recurrent Miscarriage](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
IL17A	rs2275913	AA	A	0.33	Increased relative risk of recurrent miscarriage
IL17A	rs763780	TT	C	0.05	Increased relative risk of recurrent miscarriage
IL1B	rs1143634	GG	A	0.22	AA: Decreased risk of recurrent miscarriage
IL1B	rs16944	AG	G	0.63	Increased relative risk of recurrent miscarriage
TNF	rs1800629	GG	A	0.15	higher TNF-alpha; increased risk of recurrent miscarriages
IL10	rs1800871	AG	G	0.72	Decreased risk of recurrent miscarriages (Asian ancestry)
IL10	rs1800896	TT	C	0.45	CC: Increased risk of recurrent miscarriage in Caucasians

[Article: Twins](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
FSHB	rs11031006	--	A	0.13	Lower odds of having twins
SMAD3	rs17293443	--	C	0.21	Increased possibility of fraternal twins

[Article: MTHFR and Pregnancy](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
MTHFR	rs1801133	GG	A	0.33	C677T allele, reduced MTHFR function, increased risk of miscarriages and IVF failure
MTHFR	rs1801131	GT	G	0.3	A1298C allele, slightly reduced MTHFR function, increased risk of miscarriages for GG
MTHFR	rs2274976	CC	T	0.04	decreased MTHFR efficiency; 3-fold increased risk of cleft lip; increased risk of neural tube defect
MTHFR	rs868014	--	A	0.02	A/A: increased risk of low ovarian reserve

[Article: Egg Quality](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
SLC30A2	rs587776926	--	C	0	carrier of a zinc transporter deficiency mutation
SLC30A2	rs185398527	--	T	0.0001	carrier of a zinc transporter deficiency mutation
SLC30A2	rs117153535	CC	A	0.0002	lower zinc levels in breast milk
MTNR1B	rs10830963	CC	G	0.27	increased risk of PCOS
MTHFR	rs1801133	GG	A	0.33	C677T variant, increased need for folate
MTHFR	rs1801131	GT	G	0.3	A1298C variant, slightly increased need for folate
GSTP1	rs1695	AA	G	0.32	G/G: reduced detoxification via GSTP1
GSTM1	rs366631	--	A	0.78	AA: GSTM1 deletion; decreased mercury detoxification
ADAMTS19	rs246246	--	G	0.07	linked to premature ovarian insufficiency
ESR1	rs9340799	GG	G	0.33	decreased risk of premature ovarian insufficiency (good)
HLAF	rs2523393	AA	G	0.42	slower average time to pregnancy

[🔗 Article: Nutrigenomics and pregnancy.](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
PEMT	rs7946	CC	T	0.68	decreased PEMT enzyme activity; increased relative risk of preterm birth with low choline intake
PEMT	rs12325817	--	G	0.24	increased risk of organ dysfunction with low choline diet, lower betaine levels in pregnant women with inadequate choline intake
BHMT	rs3733890	GG	A	0.3	reduced BHMT, reduced conversion of choline to betaine, increased risk of early-onset heart disease with poor diet
BHMT	rs651852	TT	C	0.55	CC: increased risk of cleft lip, an indicator of low methyl groups
MTHFR	rs1801133	GG	A	0.33	increased relative risk of preterm birth or low birth weight with low folate; increased risk of neural tube defects with low folate intake; increased relative risk of preeclampsia
MTHFR	rs1801131	GT	G	0.3	increased relative risk of small for gestational age baby without adequate maternal folate in pregnancy
MTHFR	rs2274976	CC	T	0.04	decreased MTHFR efficiency; 3-fold increased risk of cleft lip, increased risk of neural tube defect
MTHFR	rs17367504	AA	G	0.14	protective against hypertension, reduced risk of preeclampsia
MTHFD1	rs2236225	AA	A	0.43	decreased MTHFD1 enzyme stability, more of a reliance on choline as a methyl donor, increased risk of congenital heart disease in offspring
FADS1	rs174546	CC	T	0.32	low FADS1 enzyme activity; more likely to have low DHA and EPA levels during pregnancy if not supplementing
FADS2	rs1535	AA	G	0.32	low FADS2 enzyme activity, decreased ALA to EPA conversion, likely to have low DHA and EPA levels during pregnancy if not supplementing
GSTP1	rs1695	AA	G	0.32	reduced function, lower mental development scores in children when maternal diet has a high intake of fish containing mercury
TMPRSS6	rs855791	GA	A	0.43	AA: 4-fold increased relative risk of iron-deficient anemia in pregnancy with low iron intake
HFE	rs1800562	--	A	0.05	C282Y variant, most common cause of hereditary hemochromatosis, highest ferritin levels; lower serum manganese levels
HFE	rs1799945	CC	G	0.13	H63D variant, increase the risk of (mild) hemochromatosis, and increased ferritin levels; linked to lower serum manganese levels
CBS	rs2851391	CT	T	0.43	TT: increased risk of congenital heart disease in baby with poor maternal diet
ADIPOQ	rs2241766	TT	G	0.11	GG: increased risk of congenital heart disease in baby with poor maternal diet
ANRIL	rs4977574	GG	G	0.48	lower risk of preeclampsia
ANRIL	rs1333048	CC	C	0.49	lower risk of preeclampsia
LRP8	rs2297660	--	T	0.38	in women with preeclampsia, reduced risk of low birth weight baby

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP11B2	rs1799998	GA	G	0.42	increased relative risk of hypertension in many population groups; higher relative risk of preeclampsia
CBS	rs234714	CC	T	0.23	TT: increased risk of congenital heart disease in baby with poor maternal diet

Detoxification and Fertility:

[Article: SOD1](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
SOD1	rs1041740	CC	T	0.3	Decreased SOD1; increased risk of kidney problems with diabetes
SOD1	rs2070424	AA	G	0.08	Increased SOD1; protective against Alzheimer's and hearing loss
SOD1	i5000864	--	A	0	Rare mutations linked to ALS
SOD1	i5048521	--	C	0	Rare mutations linked to ALS
SOD1	rs10432782	--	G	0.07	Higher SOD1 enzyme activity, greater risk of noise-induced hearing loss
SOD1	rs121912436	--	A	0	Rare mutations linked to ALS
SOD1	rs121912431	--	C	0	Rare mutations linked to ALS

[Article: Phthalate Detoxification](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
GSTP1	rs1695	AA	G	0.32	Reduced function; slower elimination of phthalates
SOD2	rs5746136	CT	T	0.28	Higher phthalate levels; increased asthma risk
CYP2B6	rs3745274	TT	T	0.25	TT: decreased CYP2B6 enzyme needed for eliminating phthalates
GSTM1	rs366631	--	A	0.78	AA: deletion (null) GSTM1 gene; Women with phthalate exposure are at a 5-fold increased risk of fibroids.
HSPA1L	rs2227956	AA	G	0.15	G/G: decreased HSP70; insulin resistance not tied to phthalate exposure
CYP2C9	rs1799853	CC	T	0.1	decreased breakdown and clearance of phthalates
CYP2C8	rs11572080	CC	T	0.08	decreased breakdown and clearance of phthalates

[Article: BPA Detoxification](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
COX2	rs5277	--	C	0.88	C/C: increase in risk for liver dysfunction with BPA exposure
CAT	rs769217	CC	T	0.23	Increased risk of liver dysfunction in the elderly with BPA exposure
SOD2	rs4880	AA	A	0.51	Increased risk of liver dysfunction with BPA exposure
UGT2B15	rs1902023	AA	A	0.5	AA: decreased enzyme activity
UGT1A1	rs34983651	--	I	0	Decreased liver clearance of BPA
SULT1A1	i6018900	--	T	0.31	low activity of the enzyme
SULT1A1	rs9282861	--	T	0.31	low activity of the enzyme
ABCB4	rs2230028	CT	C	0.08	reduced ABCB4 activity, increased gallstone risk
CYP17A1	rs743572	GG	G	0.39	higher CYP17A1 function (possibly higher sex hormone levels); increased risk of prostate cancer, increased risk of PCOS, increased risk of acne vulgaris, and decreased risk of breast cancer in certain situations

[Article: Glyphosate \(RoundUp\) Detoxification](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP1A1	rs1048943	TT	C	0.04	More likely to have low acetylcholinesterase with glyphosate exposure.

Disclaimer: The information provided in the report and by Genetic Lifehacks is for informational purposes only. It is not intended to diagnose or treat any disease. Please visit www.GeneticLifehacks.com for more detailed information and full references.